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2027 CYCLE

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CUET UG

Chemistry

Alcohols, Phenols and Ethers

Official Syllabus Unit VII · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus, the current 2027 baseline, Unit VII)	Skills the questions test
Classification & Structures	Classification of Alcohols, Phenols and Ethers (mono-, di-, tri- or polyhydric; 1°/2°/3°), IUPAC nomenclature, electronic structures of functional groups ($C-O-H$ and $C-O-C$ bond angles).	Classifying hydroxy/alkoxy compounds, assigning IUPAC names, explaining why the $C-O-C$ angle in ethers exceeds the tetrahedral angle (111.7°).
Preparation of Alcohols & Phenols	Methods of preparation of Alcohols (hydration of alkenes, hydroboration-oxidation, reduction of carbonyls and esters, Grignard additions) and Phenols (from haloarenes, benzenesulphonic acid, diazonium salts, and cumene).	Applying Markovnikov vs anti-Markovnikov syn addition, distinguishing $NaBH_4$ vs $LiAlH_4$ selectivity, and detailing the industrial cumene-to-phenol process.
Physical & Chemical Properties	Physical properties (boiling points, intermolecular hydrogen bonding, solubility). Chemical reactions of alcohols (acidity, esterification, reactions involving $C-O$ cleavage, Lucas test, Victor Meyer test, dehydrogenation over Cu, dehydration).	Ordering boiling points and solubilities, predicting Lucas turbidity times, calculating esterification rates, and identifying dehydration alkene products.
Chemistry of Phenols	Reactivity of Phenols: acidity and substituent effects ($-NO_2$ vs $-CH_3$), electrophilic aromatic substitutions (halogenation, nitration), Kolbe's reaction, Reimer-Tiemann reaction, reaction with zinc dust, oxidation.	Explaining phenoxide resonance stability, separating o- and p-nitrophenol by steam distillation, deducing dichlorocarbene intermediate, and synthesizing aspirin.
Ethers: Synthesis & Cleavage	Methods of preparation of Ethers (acid-catalysed dehydration of alcohols, Williamson ether synthesis), physical properties, chemical reactions (cleavage of $C-O$ bond by hydrogen halides, electrophilic substitution of anisole).	Executing Williamson syntheses with primary halides vs avoiding 3° elimination, predicting HI cleavage products (S_N2 vs S_N1), and directing anisole substitutions.
Commercial Alcohols	Manufacture, uses, and physiological hazards of Methanol (CH_3 OH) and Ethanol (C_2H_5 OH), denatured alcohol.	Detailing water gas catalytic synthesis of methanol, yeast enzyme fermentation steps (invertase and zymase), and physiological toxicity mechanisms.

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EXAM PATTERN

CUET UG Paper Pattern & Weightage: The Chemistry paper carries **50 compulsory questions** in **60 minutes** (+5 marks for correct, −1 mark for incorrect). Unit VII (Alcohols, Phenols and Ethers) contributes 4 to 6 questions focusing on Lucas & Victor Meyer distinctions, Williamson synthesis with 3° halides, Kolbe and Reimer-Tiemann mechanisms, ether cleavage with HI, and phenol substituent acidity trends.

Module 1: Classification, Nomenclature & Structural Geometry

1. Classification & Nomenclature

1. **Alcohols:**
   - **By Hydroxyl Group Count:** Monohydric (ethanol), Dihydric / Glycols (ethane-1,2-diol), Trihydric / Glycerols (propane-1,2,3-triol), and Polyhydric (sorbitol, mannitol).
   - **By Carbon Hybridisation ($C(sp^3) - OH$):**
     - **Primary (1°):** $-OH$ attached to $-CH_2-$ (e.g. ethanol, propan-1-ol).
     - **Secondary (2°):** $-OH$ attached to $>CH-$ (e.g. propan-2-ol, butan-2-ol).
     - **Tertiary (3°):** $-OH$ attached to $>C-$ (e.g. 2-methylpropan-2-ol).
     - **Allylic Alcohols:** $-OH$ bonded to sp^3 carbon adjacent to $C=C$ ($CH_2=CH-CH_2$ OH).
     - **Benzylic Alcohols:** $-OH$ bonded to sp^3 carbon adjacent to benzene ring ($C_6H_5CH_2$ OH).
   - **Vinylic Alcohols ($C(sp^2) - OH$):** $-OH$ attached directly to $C=C$ (e.g. vinyl alcohol, $CH_2=$ CH-OH, which rapidly tautomerises to acetaldehyde).
2. **Phenols:** Hydroxyl group is bonded directly to an aromatic benzene ring (Ar-OH).
   - Monohydric: Phenol (C_6H_5 OH), o-cresol (2-methylphenol), m-cresol, p-cresol.
   - Dihydric: Catechol (benzene-1,2-diol), Resorcinol (benzene-1,3-diol), Quinol / Hydroquinone (benzene-1,4-diol).
   - Trihydric: Pyrogallol (benzene-1,2,3-triol), Hydroxyquinol (benzene-1,2,4-triol), Phloroglucinol (benzene-1,3,5-triol).
3. **Ethers:** Compounds containing an alkoxy or aryloxy group ($R-O-R'$ or $R-O-$Ar).
   - **Symmetrical / Simple Ethers:** Both alkyl groups are identical (e.g. diethyl ether, $C_2H_5-O-C_2H_5$).
   - **Unsymmetrical / Mixed Ethers:** Alkyl/aryl groups differ (e.g. ethyl methyl ether, $C_2H_5-O-CH_3$; anisole, $C_6H_5-O-CH_3$).

2. Molecular Geometry & Bond Angle Comparison

In alcohols, phenols, and ethers, the oxygen atom is sp^3-hybridised with two bonding pairs and two non-bonding lone pairs:

- **Methanol ($C-O-H$):** The bond angle is 108.9°, slightly smaller than the ideal tetrahedral angle (109.5°). This slight compression is caused by the greater electrostatic repulsion exerted by the two unshared lone pairs on oxygen against the bonding pairs (**lone pair - bonding pair repulsion**).
- **Phenol ($C-O-H$):** The bond angle is 109°. The $C-O$ bond length in phenol (136 pm) is substantially shorter than in methanol (142 pm). This shortening arises from:
  1. **Resonance Effect:** Partial double-bond character between ring carbon and oxygen.
  2. **Hybridisation Difference:** Carbon is sp^2-hybridised (33% s-character), holding electrons tighter.
- **Dimethyl Ether ($C-O-C$):** The bond angle is 111.7°, significantly **larger than the tetrahedral angle** (109.5°). **Critical Examination Fact:** The repulsive steric interaction between the two bulky, electron-rich alkyl groups outweighs the lone-pair compression, forcing the $C-O-C$ bond angle to widen to 111.7°.

Module 2: Preparation of Alcohols and Phenols

1. Synthesis of Alcohols

1. **From Alkenes:**
   - **Acid-Catalysed Hydration:** Direct addition of water in the presence of dilute mineral acid (H_2SO_4):

$$CH_3 - CH = CH_2 + H_2O \xrightarrow{H^+} CH_3 - CH(OH) - CH_3 \quad \text{(Markovnikov addition)}$$

Mechanism: Stepwise addition via a planar carbocation intermediate. Prone to 1,2-hydride or 1,2-methyl shifts to yield rearranged alcohols.

- **Hydroboration-Oxidation (H.C. Brown, Nobel Prize 1979):** Diborane (B_2H_6) reacts with alkenes to form trialkylboranes, which are oxidized by alkaline hydrogen peroxide (H_2O_2/OH^-):

$$3CH_3 - CH = CH_2 + (BH_3)_2 \rightarrow (CH_3CH_2CH_2)_3B \xrightarrow{H_2O_2/OH^-} 3CH_3CH_2CH_2OH + B(OH)_3$$

CUET Hallmark Rule: Overall results in **anti-Markovnikov addition of water** via concerted **syn-addition**. Because no carbocation is formed, **no structural rearrangements occur**.

2. **From Carbonyl Compounds:**
   - **Reduction of Aldehydes and Ketones:**
     - Aldehydes are reduced to primary (1°) alcohols; ketones are reduced to secondary (2°) alcohols.
     - Reagents: Sodium borohydride ($NaBH_4$) in ethanol or Lithium aluminium hydride ($LiAlH_4$) in dry ether.
   - **Reduction of Carboxylic Acids & Esters:**
     - Carboxylic acids are reduced exclusively to 1° alcohols by the powerful reducing agent $LiAlH_4$.
     - Commercially, acids are converted to esters and reduced via catalytic hydrogenation (**Bouveault-Blanc Reduction**):

$$R - COOH \xrightarrow{R'OH\ /\ H^+} R - COOR' \xrightarrow[\text{Catalyst}]{H_2} R - CH_2OH + R' - OH$$

3. **From Grignard Reagents (Carbon-Carbon Chain Extension):** Nucleophilic addition of $R - MgX$ to carbonyl groups followed by acidic hydrolysis:
   - **Methanal (Formaldehyde, HCHO):** Yields **Primary (1°) Alcohols**:

$$HCHO + R - MgX \rightarrow R - CH_2OMgX \xrightarrow{H_2O} R - CH_2OH + Mg(OH)X$$

   - **Other Aldehydes ($R' - CHO$):** Yields **Secondary (2°) Alcohols**:

$$R' - CHO + R - MgX \rightarrow R' - CH(OMgX) - R \xrightarrow{H_2O} R' - CH(OH) - R$$

   - **Ketones (R'_2 C=O):** Yields **Tertiary (3°) Alcohols**:

$$R'_2\ C{=}O + R - MgX \rightarrow R'_2C(OMgX) - R \xrightarrow{H_2O} R'_2C(OH) - R$$

   - **Oxymercuration-Demercuration (Markovnikov Hydration without Rearrangement):** When alkenes react with mercuric acetate [$Hg(OAc)_2$] in aqueous THF followed by reduction with sodium borohydride ($NaBH_4$ in basic medium), an alcohol is produced:

$$CH_3 - CH = CH_2 + Hg(OAc)_2 + H_2O \rightarrow CH_3 - CH(OH) - CH_2 - HgOAc \xrightarrow{NaBH_4/OH^-} CH_3 - CH(OH) - CH_3$$

CUET Comparison Insight: Unlike acid-catalysed hydration, oxymercuration proceeds via a bridged cyclic mercurinium ion intermediate. Because free carbocations are never formed, **zero carbocation rearrangement** occurs, delivering pure Markovnikov addition.

   - **Grignard Addition to Oxirane (Ethylene Oxide) — Chain Lengthening by Two Carbons:** Reaction of Grignard reagents ($R - MgX$) with cyclic ethylene oxide (oxirane) followed by acidic hydrolysis produces **primary (1°) alcohols lengthened by exactly two carbon atoms**:

$$R - MgX + \underbrace{H_2C - CH_2}_{\text{Oxirane}}/O \rightarrow R - CH_2CH_2 - OMgX \xrightarrow{H_2O} R - CH_2CH_2 - OH + Mg(OH)X$$

Exam Strategy: If asked to prepare propan-1-ol from methyl bromide, use methylmagnesium bromide with oxirane!

2. Industrial & Laboratory Synthesis of Phenols

1. **From Cumene (Isopropylbenzene — Dominant Commercial Route):** Over 90% of global synthetic phenol is manufactured via this single elegant process:
   - Cumene is oxidized by passing air or oxygen through it at $368 - 408$ K to produce **cumene hydroperoxide**.
   - Cumene hydroperoxide is treated with dilute sulphuric acid (H_2SO_4) at $323 - 363$ K:

$$C_6H_5 - CH(CH_3)_2 + O_2 \rightarrow C_6H_5 - C(CH_3)_2 - O\text{-}O\text{-}H \xrightarrow{H^+} C_6H_5\ OH + CH_3COCH_3$$

   Industrial Merit: Generates equimolar quantities of **Acetone (propan-2-one)** as a high-value commercial byproduct.
2. **From Haloarenes (Dow's Process):** Chlorobenzene is heated with $6 - 8\%$ aqueous sodium hydroxide at 623 K under 300 atm pressure, followed by acidification:

$$C_6H_5\ Cl + 2\ NaOH \xrightarrow{623\ K, 300\ atm} C_6H_5\ ONa + NaCl + H_2\ O \xrightarrow{HCl} C_6H_5\ OH$$

3. **From Benzenediazonium Salts:** Benzenediazonium chloride is warmed gently with water or dilute mineral acid:

$$C_6H_5 - N_2^+Cl^- + H_2\ O \xrightarrow{warm} C_6H_5\ OH + N_2 \uparrow + HCl$$

4. **From Benzenesulphonic Acid:** Benzene is sulphonated with oleum; sodium benzenesulphonate is fused with solid sodium hydroxide at 573 K, yielding sodium phenoxide which on acidification gives phenol.

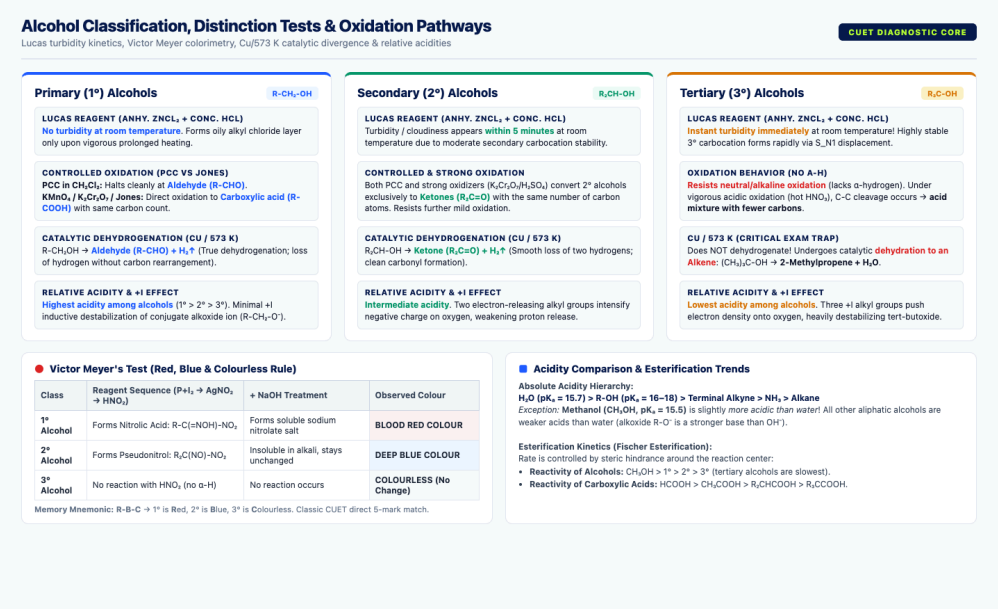


Figure 1: Alcohol classification, diagnostic distinction tests (Lucas turbidity kinetics & Victor Meyer colorimetry), Cu/573 K catalytic divergence, and relative acidities.

Module 3: Physical Properties & Distinction Tests of Alcohols

1. Physical Properties Trends

- **Boiling Points:** Alcohols have strikingly higher boiling points than alkanes, haloalkanes, or ethers of comparable molecular mass:

Ethanol ($M = 46$, b.p. $= 351$ K) versus Propane ($M = 44$, b.p. $= 231$ K) versus Methoxymethane ($M = 46$, b.p. $= 248$ K)

  Explanation: Intermolecular hydrogen bonding ($O - H \cdots O$) links alcohol molecules into extensive dynamic polymeric aggregates, requiring substantial thermal energy to vaporise.
- **Effect of Branching on Isomers:** Boiling point decreases with chain branching:

n-butyl alcohol (391 K) > isobutyl alcohol (381 K) > sec-butyl alcohol (373 K) > tert-butyl alcohol (356 K)

Branching diminishes surface area, making the molecule more compact and weakening van der Waals dispersion forces.

- **Water Solubility:** Lower alcohols (methanol, ethanol, propanol) are completely miscible with water in all proportions because their hydroxyl groups form strong hydrogen bonds with water molecules. Solubility drops sharply as the non-polar, hydrophobic hydrocarbon chain extends (C_5 and higher).

2. Chemical Properties & Acidity of Alcohols

Alcohols act as versatile amphoteric compounds (both weak Brønsted acids and Lewis bases):

1. **Acidity Hierarchy:**

$$\mathrm{H_2O}\ (pK_a = 15.7) > R-\mathrm{OH}\ (pK_a = 16-18) > \text{Terminal Alkynes} > \mathrm{NH_3} > \text{Alkanes}$$

Crucial Exception: Methanol ($\mathbf{CH_3OH}$, $\boldsymbol{pK_a = 15.5}$) is slightly **more acidic than water**! All other aliphatic alcohols are weaker acids than water. Consequently, alkoxide ions ($R-O^-$) are stronger bases than hydroxide ion (OH^-).

2. **Acidity Order Among Alcohol Classes:**

$$1^\circ\ \text{Alcohol} > 2^\circ\ \text{Alcohol} > 3^\circ\ \text{Alcohol}$$

Mechanism: Alkyl groups are electron-releasing ($+I$ inductive effect). In tertiary alcohols, three alkyl groups pump electron density onto oxygen, intensifying the negative charge on the conjugate alkoxide ion and heavily destabilizing it.

3. **Esterification (Fischer Esterification):** Reaction of alcohols with carboxylic acids in the presence of concentrated H_2SO_4:

$$R-\mathrm{COOH} + R'-\mathrm{OH} \overset{\mathrm{H^+}}{\leftrightarrow} R-\mathrm{COOR'} + \mathrm{H_2O}$$

   - **Steric Factor Dictates Rate:** Reactivity of alcohols follows: $CH_3OH > 1^\circ > 2^\circ > 3^\circ$. Bulkier tertiary alcohols esterify sluggishly.

4. **Mechanism of Acid-Catalysed Dehydration of Alcohols (E_1 Pathway):** Dehydration of alcohols to form alkenes proceeds via an E_1 elimination mechanism:
   - **Step 1 (Fast Protonation):** Formation of protonated alcohol (alkyloxonium ion):

$$\mathrm{C_2H_5OH} + \mathrm{H^+} \leftrightarrow \mathrm{C_2H_5OH_2^+}$$

   - **Step 2 (Slow & Rate-Determining):** Cleavage of $C-O$ bond and loss of water to generate a planar carbocation:

$$\mathrm{C_2H_5OH_2^+} \overset{\text{slow}}{\rightarrow} \mathrm{CH_3-CH_2^+} + \mathrm{H_2O}$$

   Ease of Dehydration: Follows carbocation stability: 3° alcohol > 2° alcohol > 1° alcohol. Tertiary alcohols dehydrate under mild conditions (20%H_3PO_4 at 358 K), whereas primary alcohols require concentrated H_2SO_4 at 443 K.
   - **Step 3 (Fast Deprotonation):** Elimination of a β-hydrogen proton to form the double bond:

$$\mathrm{CH_3-CH_2^+} \rightarrow \mathrm{CH_2=CH_2} + \mathrm{H^+}$$

   Saytzeff's Rule: When multiple alkenes can form, the more highly substituted, thermodynamically stable alkene predominates (e.g., butan-2-ol yields but-2-ene as 80% major product).

5. **Mechanism of Fischer Esterification & Isotopic Evidence:** When an alcohol is refluxed with a carboxylic acid in the presence of concentrated H_2SO_4, esterification proceeds through protonation of the carboxyl carbonyl oxygen, nucleophilic addition of the alcohol oxygen, proton transfer, elimination of water, and loss of proton. **Isotopic Tracer Proof:** By conducting esterification with O-18 labelled methanol ($CH_3-{}^{18}O-H$), the heavy isotope is found exclusively in the **ester**, proving that the $C-OH$ bond of the **acid** breaks, while the $O-H$ bond of the **alcohol** cleaves!

3. Diagnostic Chemical Distinction Tests

1. **The Lucas Test (Distinction via $C-O$ Cleavage Kinetics):** Uses Lucas Reagent (concentrated HCl and anhydrous $ZnCl_2$). Alcohols convert into insoluble alkyl chlorides, creating turbidity/ cloudiness:
   - **Tertiary (3°) Alcohols:** Produce instant turbidity immediately at room temperature (forms stable 3° carbocation via S_N1).
   - **Secondary (2°) Alcohols:** Turbidity appears within 5 minutes at room temperature.
   - **Primary (1°) Alcohols:** Remain clear at room temperature; turbidity appears only upon prolonged heating.
2. **Victor Meyer's Test (Distinction via Colorimetry):** Sequence: Alcohol $+\frac{P}{I_2} \rightarrow$ Iodoalkane $+AgNO_2 \rightarrow$ Nitroalkane $+HNO_2 \rightarrow$ Alkali (NaOH):
   - **Primary (1°) Alcohol:** Forms nitrolic acid, which dissolves in NaOH to yield an intense **Blood Red colour**.
   - **Secondary (2°) Alcohol:** Forms pseudonitrol, which is insoluble in alkali and produces a **Deep Blue colour**.
   - **Tertiary (3°) Alcohol:** Lacks α-hydrogen; does not react with nitrous acid, remaining completely **Colourless**.

   CUET Memory Formula: R-B-C (1° = Red, 2° = Blue, 3° = Colourless).
3. **Catalytic Dehydrogenation over Heated Copper ($\frac{Cu}{573}$ K):**
   - **Primary (1°) Alcohol:** Dehydrogenated cleanly to an **Aldehyde**:

$$R-CH_2\ OH \xrightarrow{\frac{Cu}{573}\ K} R-CHO + H_2 \uparrow$$

   - **Secondary (2°) Alcohol:** Dehydrogenated to a **Ketone**:

$$R_2\ CH\text{-}OH \xrightarrow{\frac{Cu}{573}\ K} R_2\ C{=}O + H_2 \uparrow$$

   - **Tertiary (3°) Alcohol (HIGH-YIELD TRAP):** Because it lacks an α-hydrogen atom, it **cannot dehydrogenate**. Instead, it undergoes catalytic **dehydration to an Alkene**:

$$(CH_3)_3C-OH \xrightarrow{\frac{Cu}{573}\ K} (CH_3)_2C=CH_2 + H_2\ O \quad \text{(2-methylpropene)}$$

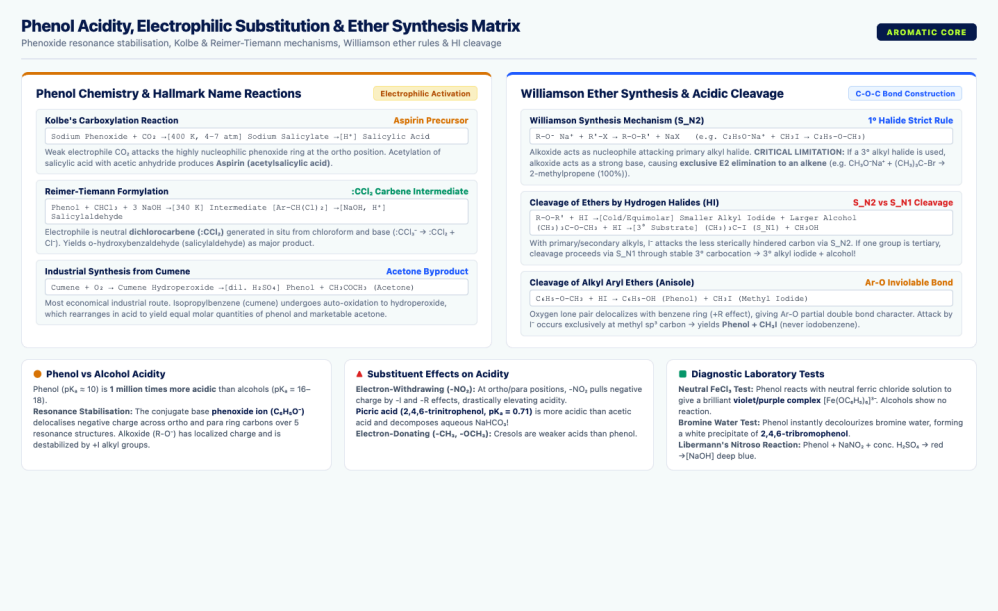


Figure 2: Phenols acidity mechanisms, phenoxide resonance stabilization, hallmark name reactions (Kolbe, Reimer-Tiemann), Williamson ether synthesis constraints, and HI cleavage regiochemistry.

Module 4: Chemical Properties & Name Reactions of Phenols

1. Acidity of Phenols & Substituent Effects

Phenol ($pK_a \approx 10$) is an acidic organic compound—approximately **one million times more acidic than aliphatic alcohols** ($pK_a \approx 16-18$):

- **Phenoxide Ion Resonance Stabilisation:** Ionisation yields a phenoxide ion ($C_6H_5O^-$). The negative charge is delocalised over the ortho and para positions of the aromatic ring across five contributing resonance structures. In contrast, the alkoxide ion ($R-O^-$) has its negative charge localized entirely on oxygen and is destabilized by $+I$ alkyl groups.
- **Phenol vs Carboxylic Acid:** Although phenol turns blue litmus red and dissolves in aqueous NaOH, it is a weaker acid than carbonic acid (H_2CO_3) and carboxylic acids ($R-COOH$). Hence, **phenol does NOT react with sodium bicarbonate ($NaHCO_3$)** and produces no effervescence of CO_2.
- **Substituent Influence on Phenol Acidity:**
  - **Electron-Withdrawing Groups ($-NO_2, -CN$, Halogens):** Disperse negative charge from the phenoxide ring via $-I$ and $-R$ effects, dramatically increasing acidity. The activating effect is maximal at **ortho** and **para** positions:

ic acid) ($pK_a = 0.71$) > 2,4-dinitrophenol ($pK_a = 4.0$) > p-nitrophenol ($pK_a = 7.15$) > o-nitrophenol ($pK_a = 7.23$) > m-nitrophenol (

Examination Fact: Picric acid is so strongly acidic that it decomposes aqueous $NaHCO_3$, releasing effervescence of CO_2!

  - **Electron-Donating Groups ($-CH_3, -OCH_3, -NH_2$):** Intensify negative charge on the phenoxide ring via $+I$ and hyperconjugation/resonance, decreasing acidity:

Phenol ($pK_a = 10.0$) > p-cresol ($pK_a = 10.26$) > m-cresol ($pK_a = 10.08$) > o-cresol ($pK_a = 10.28$)

2. Electrophilic Aromatic Substitution of Phenols

The hydroxyl group is powerfully activating and strictly **ortho/para-directing**:

1. **Halogenation in Different Solvents:**
   - **In Non-Polar Solvent (CS_2 or $CHCl_3$ at 273 K):** Phenol is minimally ionised; monobromination yields **p-bromophenol (major)** and o-bromophenol (minor):

$$C_6H_5OH + Br_2 \xrightarrow{CS_2, 273\ K} \text{p-bromophenol (80\%)} + \text{o-bromophenol (20\%)}$$

   - **In Aqueous Medium ($\frac{Br_2}{H_2}O$):** Water ionises phenol completely into phenoxide ion ($C_6H_5O^-$), which hyperactivates the ring towards electrophilic attack. Instantaneous tribromination occurs, yielding a characteristic **white precipitate of 2,4,6-tribromophenol**:

$$C_6H_5OH + 3Br_2(aq) \rightarrow \text{2,4,6-tribromophenol} \downarrow \quad \text{(White Precipitate)} + 3\ HBr$$

2. **Nitration of Phenol:**
   - **With Dilute Nitric Acid (dil. HNO_3 at 298 K):** Yields a mixture of ortho-nitrophenol and para-nitrophenol:
     - **o-Nitrophenol:** Features **intramolecular hydrogen bonding** (chelation within the same molecule). It has a lower boiling point and is **steam volatile**.
     - **p-Nitrophenol:** Features **intermolecular hydrogen bonding** (linking separate molecules into aggregates). It has a higher boiling point and is **non-steam volatile**.

     Industrial Separation: The two isomers are cleanly separated by **steam distillation**.
   - **With Concentrated Nitric Acid ($\frac{\text{conc. } HNO_3}{\text{conc. } H_2}SO_4$):** Direct nitration produces **2,4,6-trinitrophenol (Picric acid)** in poor yields due to destructive oxidation of the ring.

3. Hallmark Name Reactions of Phenols

1. **Kolbe's Reaction (Synthesis of Salicylic Acid):** Phenoxide ion is treated with carbon dioxide (CO_2) at 400 K under $4-7$ atm pressure, followed by acidification:

$$C_6H_5\ ONa + CO_2 \xrightarrow{400\ K, 4-7\ atm} \text{o-hydroxybenzoic acid (Salicylic acid)}$$

Pharmaceutical Gateway: Acetylation of salicylic acid with acetic anhydride in the presence of acid yields **Aspirin (acetylsalicylic acid)**:

$$\text{Salicylic Acid} + (CH_3\ CO)_2\ O \xrightarrow{H^+} \text{Acetylsalicylic acid (Aspirin)} + CH_3\ COOH$$

2. **Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde):** Heating phenol with chloroform ($CHCl_3$) in aqueous sodium hydroxide at 340 K followed by acidic hydrolysis:

$$C_6H_5\ OH + CHCl_3 + 3\ NaOH \xrightarrow{340\ K} \text{Salicylaldehyde (o-hydroxybenzaldehyde)} + 3\ NaCl + 2H_2\ O$$

Crucial Mechanistic Intermediates: The reactive electrophile is neutral **dichlorocarbene (:CCl_2)** generated in situ via α-elimination from chloroform:

$$CHCl_3 + OH^- \leftrightarrow :CCl_3^- + H_2\ O \rightarrow :CCl_2 + Cl^-$$

If carbon tetrachloride (CCl_4) is used instead of chloroform, the reaction yields **Salicylic acid**.

3. **Reaction with Zinc Dust:** Distillation of phenol with zinc dust strips the oxygen, reducing it to **benzene**:

$$C_6H_5\ OH + Zn \xrightarrow{\Delta} C_6H_5 - H + ZnO$$

4. **Oxidation with Chromic Acid ($Na_2Cr_2\frac{O_7}{H_2}SO_4$):** Conjugated oxidation generates **p-benzoquinone**, a bright yellow conjugated diketone.
5. **Neutral Ferric Chloride Test ($FeCl_3$):** Phenols react with neutral $FeCl_3$ to form intensely coloured violet/purple coordination complexes: $\left[Fe(OC_6H_5)_6\right]^{3-}$. Aliphatic alcohols give no colour.

Module 5: Ethers — Synthesis, Properties & Acidic Cleavage

1. Methods of Preparation of Ethers

1. **Dehydration of Alcohols (Temperature-Controlled Divergence):** When ethanol is heated with concentrated sulphuric acid, the reaction outcome is controlled by temperature:
   - **At 413 K (140° C):** Intermolecular dehydration via bimolecular nucleophilic substitution (S_N2) yields **Diethyl ether**:

$$2C_2H_5\ OH \xrightarrow{\text{conc. } H_2SO_4, 413\ K} C_2H_5 - O - C_2H_5 + H_2\ O$$

   - **At 443 K (170° C):** Intramolecular dehydration via β-elimination dominates, yielding **Ethene**:

$$C_2H_5\ OH \xrightarrow{\text{conc. } H_2SO_4, 443\ K} CH_2 = CH_2 + H_2\ O$$

2. **Williamson Ether Synthesis (Universal Laboratory Route):** An alkyl halide is heated with a sodium alkoxide or phenoxide in an S_N2 displacement:

$$R - O^-Na^+ + R' - X \rightarrow R - O - R' + NaX$$

THE SACRED WILLIAMSON LIMITATION (MUST MASTER FOR CUET):

- The alkyl halide $R' - X$ **must strictly be primary (1°)!**
- If a **tertiary (3°) alkyl halide** is used, the alkoxide behaves as a strong base rather than a nucleophile, causing **exclusive E2 dehydrohalogenation to an Alkene**:

$$CH_3O^-Na^+ + (CH_3)_3C - Br \rightarrow (CH_3)_2C = CH_2 (100\%\ \text{alkene}) + CH_3\ OH + NaBr$$

- To prepare tert-butyl methyl ether, one must react **sodium tert-butoxide with methyl bromide**:

$$(CH_3)_3C - O^-Na^+ + CH_3\ Br \rightarrow (CH_3)_3C - O - CH_3 + NaBr \quad \text{(Smooth S_N2 synthesis)}$$

2. Acidic Cleavage of Ethers by Halogen Acids ($\frac{HI}{HBr}$)

Ethers are relatively unreactive compounds, but their etherial $C - O$ linkage is cleaved under vigorous conditions by concentrated halogen acids ($HI > HBr \gg HCl$):

1. **Cleavage of Symmetrical Ethers:**

$$R-O-R+\mathrm{HI}\rightarrow R-I+R-\mathrm{OH}\xrightarrow{\text{excess HI},\Delta}2R-I+\mathrm{H_2\ O}$$

2. **Cleavage of Unsymmetrical Ethers Containing Primary / Secondary Alkyl Groups (S_N2):** When an unsymmetrical ether containing 1° or 2° alkyl groups is cleaved with equimolar cold HI, the halide ion (I^-) attacks the **less sterically hindered carbon atom**:

$$\mathrm{CH_3}-O-\mathrm{CH_2CH_2CH_3}+\mathrm{HI}\rightarrow\mathrm{CH_3}-I+\mathrm{CH_3CH_2CH_2\ OH}$$

3. **Cleavage of Ethers Containing a Tertiary Alkyl Group (S_N1):** If one of the alkyl groups is tertiary (3°), cleavage proceeds via an S_N1 mechanism through a highly stable tertiary carbocation. The halide ion attacks the tertiary carbon:

$$(\mathrm{CH_3})_3C-O-\mathrm{CH_3}+\mathrm{HI}\rightarrow(\mathrm{CH_3})_3C-I+\mathrm{CH_3\ OH}$$

4. **Cleavage of Alkyl Aryl Ethers (Anisole):** When anisole ($\mathrm{C_6H_5}-O-\mathrm{CH_3}$) is heated with concentrated HI, the products are exclusively **Phenol and Methyl Iodide**:

$$\mathrm{C_6H_5}-O-\mathrm{CH_3}+\mathrm{HI}\rightarrow\mathrm{C_6H_5\ OH}+\mathrm{CH_3}I$$

Mechanism Justification: Oxygen lone pair resonance delocalises into the aromatic ring, giving the $\mathrm{Ar}-O$ bond partial double-bond character. The $\mathrm{Ar}-O$ bond is extraordinarily strong and cannot be cleaved; nucleophilic attack occurs strictly at the sp^3 methyl carbon. Iodobenzene is never formed.

3. Electrophilic Aromatic Substitution of Alkyl Aryl Ethers (Anisole)

The methoxy group ($-\mathrm{OCH_3}$) in anisole is activating and ortho/para-directing due to $+R$ resonance delocalisation of the oxygen lone pair into the aromatic ring. Due to steric hindrance of the bulky methoxy group, the **para isomer is overwhelmingly the major product**:

1. **Halogenation:** Anisole undergoes bromination with bromine in glacial ethanoic acid, **without requiring any Lewis acid catalyst** ($\mathrm{FeBr_3}$):

$$\mathrm{C_6H_5-OCH_3}+\mathrm{Br_2}\xrightarrow{\mathrm{CH_3\ COOH}}\text{p-bromoanisole (90\% major)}+\text{o-bromoanisole (10\% minor)}$$

2. **Nitration:** Reaction with a mixture of concentrated sulphuric and nitric acids yields a mixture of ortho- and para-nitroanisole:

$$\mathrm{C_6H_5-OCH_3}+\mathrm{HNO_3}\xrightarrow{\text{conc. }\mathrm{H_2SO_4}}\text{4-nitroanisole (major)}+\text{2-nitroanisole (minor)}$$

3. **Friedel-Crafts Alkylation & Acylation:**
   - **Alkylation:** Anisole $+\mathrm{CH_3\ Cl}\xrightarrow{\text{anhyd. }\mathrm{AlCl_3}}$ 4-methoxytoluene (major) + 2-methoxytoluene (minor)
   - **Acylation:** Anisole $+\mathrm{CH_3\ COCl}\xrightarrow{\text{anhyd. }\mathrm{AlCl_3}}$ 4-methoxyacetophenone (major) + 2-methoxyacetophenone (minor)

4. Ether Storage Hazards & Peroxide Formation

- **Spontaneous Autoxidation:** On prolonged contact with atmospheric oxygen in the presence of sunlight, diethyl ether absorbs oxygen to form unstable, highly explosive **diethyl ether hydroperoxide**:

$$\mathrm{C_2H_5}-O-\mathrm{CH_2CH_3}+\mathrm{O_2}\xrightarrow{h\nu}\mathrm{C_2H_5}-O-\mathrm{CH(OOH)}-\mathrm{CH_3}$$

- **Peroxide Detection & Prevention:** Peroxides in ether can be detected by shaking with acidified ferrous sulphate ($\mathrm{FeSO_4}$) and potassium thiocyanate (KCNS), developing a deep blood-red colour of ferric thiocyanate [$\mathrm{Fe(CNS)_3}$]. Distillation of old, ether containing peroxides can cause violent explosions. To inhibit peroxide formation, ether is stored in dark amber bottles over clean **metallic copper wire** or ferrous salts.

Module 6: Commercially Important Industrial Alcohols

1. **Methanol (Methyl Alcohol / Wood Spirit, $\mathrm{CH_3\ OH}$):**
   - Historically obtained by the destructive distillation of wood.

- **Modern Industrial Synthesis:** Catalytic reduction of carbon monoxide (water gas) with hydrogen:

$$CO + 2H_2 \xrightarrow{ZnO - Cr_2O_3, 573-673\ K, 200-300\ atm} CH_3\ OH$$

- **Physical Properties & Uses:** Colourless liquid ($b.p. = 337$ K). Used as an industrial solvent for paints and varnishes, and in the manufacture of formaldehyde (HCHO).
- **Severe Toxicology:** Ingestion of even 10 mL causes permanent retinal blindness; 30 mL causes death. In the body, alcohol dehydrogenase oxidizes methanol to formaldehyde (HCHO) and formic acid (HCOOH), which coagulates cellular protoplasm and induces acute metabolic acidosis.
- **Antidote & Clinical Toxicology:** The toxicity of methanol is mediated by formaldehyde (HCHO) and formic acid (HCOOH) generated by the hepatic enzyme **alcohol dehydrogenase**. Clinically, acute methanol poisoning is treated by intravenous infusion of **ethanol**. Ethanol has a dramatically higher substrate affinity for alcohol dehydrogenase (approx. 10-fold higher K_m preference), competitively displacing methanol from the active site and allowing intact methanol to be safely excreted through kidneys and breath without generating lethal metabolites.

2. **Ethanol (Ethyl Alcohol / Grain Alcohol, $C_2H_5\ OH$):**
   - **Manufacture by Fermentation:** Prepared from sugary molasses or starchy grains using yeast enzymes:
     - Invertase converts sucrose to glucose and fructose: $C_{12}H_{22}O_{11} + H_2\ O \xrightarrow{\text{Invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$.
     - Zymase ferments glucose/fructose into ethanol and carbon dioxide: $C_6H_{12}O_6 \xrightarrow{\text{Zymase}} 2C_2H_5\ OH + 2CO_2 \uparrow$.

     Fermentation terminates naturally when ethanol concentration reaches 14%, because higher concentrations poison the yeast enzyme.
   - **Denatured Alcohol (Methylated Spirit):** Commercial ethanol is rendered unfit for drinking by adding 5% toxic methanol, copper sulphate (imparting a warning blue colour), and pyridine (a foul-smelling liquid) to curb industrial tax evasion.

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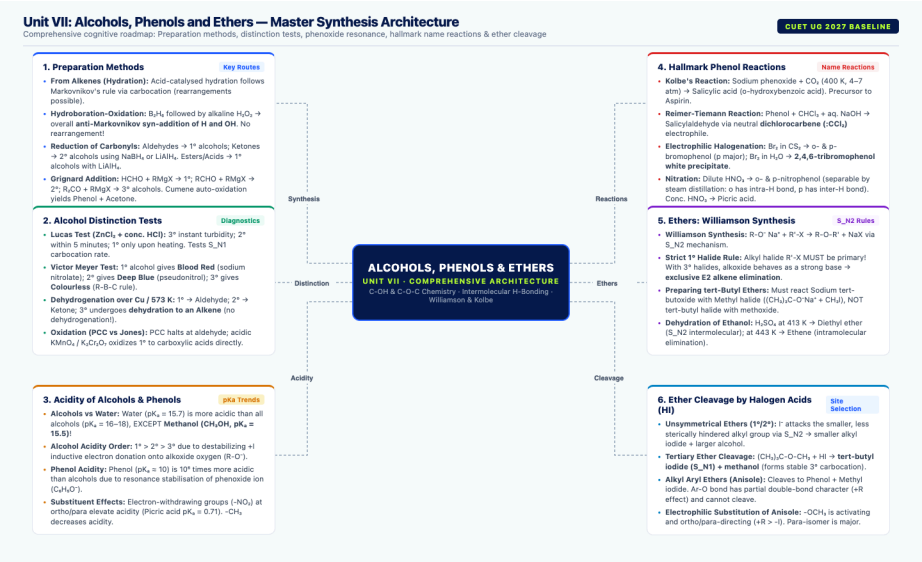


Figure 3: Master Synthesis Architecture: Complete Unit VII cognitive roadmap integrating alcohol distinctions, phenoxide resonance, hallmark name reactions, Williamson ether rules, and HI cleavage mechanisms.

MASTER SYNTHESIS MAP & STRATEGIC REACTION PATHWAYS BREAKDOWN

The cognitive architecture diagram above synthesises the core reactive transformations across Unit VII:

- **Electrophilic Attack vs Nucleophilic Substitution:** Note the stark contrast between alcohols (where $-\text{OH}$ acts as a nucleophile or is eliminated via protonation to $-\text{OH}_2^+$) and phenols (where ring activation directs electrophiles to ortho/para positions while the strong sp^2 $\text{Ar}-O$ bond remains inviolable).
- **Halogenation Regiochemistry:** Notice how water versus non-polar solvents entirely flips the bromination outcome in phenols—polar water enables phenoxide formation leading to symmetrical $2,4,6$-tribromination, whereas non-polar CS_2 preserves molecular phenol yielding predominantly p-bromophenol.
- **Dual Cleavage Pathways of Ethers:** Trace the branched logic for unsymmetrical ether cleavage with HI: sterically unhindered primary/secondary ethers proceed via bimolecular displacement (S_N2) at the smaller alkyl group, whereas tertiary ethers immediately dissociate into stable carbocations via unimolecular cleavage (S_N1).
- **Williamson Synthesis Constraints:** Always check the alkyl halide substrate first—primary halides (1°) undergo productive S_N2 etherification, whereas tertiary halides (3°) undergo 100% elimination to form alkenes due to the basicity of alkoxide ions.

UNIT VII MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Expression / Formulation	Application & CUET Problem Strategy
Ether Bond Angle	$C-O-C = 111.7^\circ > 109.5^\circ$ (Tetrahedral)	Steric repulsion between two bulky alkyl groups widens the bond angle.
Hydroboration-Oxidation	Alkene $+B_2H_6 \xrightarrow{H_2O_2/OH^-}$ Alcohol	Overall anti-Markovnikov syn addition of water. Zero carbocation rearrangements.
Lucas Test Distinction	Conc. HCl + anhy. $ZnCl_2$ turbidity kinetics	3°: Instant turbidity; 2°: 5 mins; 1°: Turbidity only on heating (S_N1).
Victor Meyer (R-B-C)	Sequence: $\frac{P}{I_2} \rightarrow AgNO_2 \rightarrow HNO_2 \rightarrow NaOH$	1° = Blood Red (nitrolic acid); 2° = Deep Blue; 3° = Colourless (no α-H).
Dehydrogenation ($\frac{Cu}{573}$ K)	$1^\circ \rightarrow$ Aldehyde; $2^\circ \rightarrow$ Ketone; $3^\circ \rightarrow$ Alkene	Tertiary alcohol dehydrates to alkene; does not dehydrogenate!
Phenol Acidity	Phenol ($pK_a \approx 10$) is $10^6 \times$ more acidic than alcohols	Phenoxide resonance-stabilized. Picric acid ($pK_a = 0.71$) gives CO_2 with $NaHCO_3$.
Kolbe's Reaction	Sodium phenoxide $+CO_2 \xrightarrow{400\ K, 4-7\ atm} \xrightarrow{H^+}$ Salicylic acid	Synthesis of o-hydroxybenzoic acid; chelation directs ortho; precursor to Aspirin.
Reimer-Tiemann Reaction	Phenol $+CHCl_3 + NaOH \xrightarrow{H^+}$ Salicylaldehyde	Electrophile is neutral dichlorocarbene ($:CCl_2$) formed via α-elimination.
Phenol Distinction Tests	$FeCl_3 \rightarrow$ Violet complex; $\frac{Br_2}{H_2O} \rightarrow$ White ppt	Neutral $FeCl_3$ gives $\left[Fe(OC_6H_5)_6\right]^{3-}$; aqueous bromine gives 2,4,6-tribromophenol.
Steam Distillation	o-nitrophenol (volatile) vs p-nitrophenol (non-volatile)	Ortho has intramolecular H-bonding; Para has intermolecular H-bonding.
Williamson Synthesis	$R-O^-Na^+ + R'-X \rightarrow R-O-R' + NaX$	$R'-X$ must be 1°! Reaction with 3° halide gives 100% alkene (E_2 elimination).
HI Ether Cleavage	Simple: I^- attacks less hindered C; 3°: forms 3° iodide	Anisole ($C_6H_5-O-CH_3$) gives Phenol $+CH_3I$ (sp^2 $Ar-O$ never cleaves).

TOP 7 HIGH-FREQUENCY CUET TRAPS & PITFALLS

1. **Dehydrogenation of Tertiary Alcohols over Heated Copper:** Primary and secondary alcohols undergo dehydrogenation to aldehydes and ketones respectively over $\frac{Cu}{573}$ K. Tertiary alcohols have no α-hydrogen and **cannot dehydrogenate**; they undergo catalytic **dehydration to an alkene** ($(CH_3)_3C-OH \rightarrow (CH_3)_2C=CH_2 + H_2O$).
2. **Lucas Test with Allylic and Benzylic Alcohols:** While ordinary primary aliphatic alcohols do not form turbidity at room temperature, **allylic** and **benzylic** alcohols yield instantaneous turbidity because their carbocations are powerfully stabilized by resonance!
3. **Acidity Comparison Between Methanol and Water:** Water ($pK_a = 15.7$) is more acidic than almost all aliphatic alcohols ($pK_a = 16-18$), **except Methanol ($pK_a = 15.5$)**. Methanol is slightly **more acidic than water**, making alkoxide ions stronger bases than hydroxide ion, except methoxide.
4. **Williamson Synthesis with Tertiary Alkyl Halides:** Reacting sodium alkoxide with a 3° alkyl halide ($(CH_3)_3C-Br + CH_3O^-Na^+$) produces **100% alkene (2-methylpropene)** via E_2 elimination, NOT an ether! To prepare tert-butyl methyl ether, use **sodium tert-butoxide and methyl bromide**.
5. **Cleavage of Anisole (Alkyl Aryl Ethers) with HI:** Heating anisole with concentrated HI yields **Phenol and Methyl Iodide**, NEVER Iodobenzene and Methanol! The sp^2 hybridized $Ar-O$ bond has partial double-bond character due to resonance and cannot undergo nucleophilic cleavage.
6. **HI Cleavage of Ethers with a Tertiary Alkyl Group:** Unsymmetrical ethers containing primary/secondary alkyl groups cleave via S_N2 where I^- attacks the less hindered carbon. If one alkyl group is tertiary (3°), cleavage proceeds via S_N1, forming the **tertiary iodide** and **primary alcohol**.
7. **Solvent Control in Phenol Bromination:** Phenol with aqueous bromine ($\frac{Br_2}{H_2O}$) gives a **white precipitate of 2,4,6-tribromophenol**, because water ionizes phenol to phenoxide which hyperactivates the ring. In non-polar CS_2 at 273 K, phenol remains unionized and yields **p-bromophenol** as major product.

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. An unknown alcohol reacts with Lucas reagent (concentrated HCl and anhydrous $ZnCl_2$) at room temperature to produce immediate turbidity. Which of the following compounds is the unknown alcohol?

(A) Butan-1-ol **(B)** Propan-2-ol
(C) 2-Methylpropan-2-ol **(D)** Ethanol

Sol. Q1. Lucas reagent reacts via an S_N1 mechanism wherein the rate is governed by carbocation stability. Tertiary alcohols form exceptionally stable 3° carbocations, undergoing immediate conversion into insoluble alkyl chlorides and producing instant turbidity at room temperature: 3° (instant) $> 2^\circ$ (5 mins) $> 1^\circ$ (heat only). 2-Methylpropan-2-ol is a 3° alcohol. **Answer: (C)**

Q2. In the Victor Meyer test for distinguishing alcohols, the colours produced after reaction with nitrous acid followed by alkalisation with sodium hydroxide for primary, secondary, and tertiary alcohols are respectively:

(A) Blood Red, Deep Blue, and Colourless **(B)** Deep Blue, Blood Red, and Colourless
(C) Colourless, Deep Blue, and Blood Red **(D)** Blood Red, Colourless, and Deep Blue

Sol. Q2. In the Victor Meyer test, primary alcohols form nitrolic acid which dissolves in alkali giving a characteristic Blood Red colour. Secondary alcohols form pseudonitrol which is alkali-insoluble and appears Deep Blue. Tertiary alcohols lack an α-hydrogen, failing to react with nitrous acid and remaining Colourless (**R-B-C Rule**). **Answer: (A)**

Q3. When vapours of 2-methylpropan-2-ol (tert-butyl alcohol) are passed over heated copper catalyst at 573 K, the major organic product obtained is:

(A) 2-Methylpropanal **(B)** Butan-2-one
(C) 2-Methylpropene **(D)** Propan-2-one

Sol. Q3. Primary and secondary alcohols undergo catalytic dehydrogenation over heated copper at 573 K to yield aldehydes and ketones respectively. However, tertiary alcohols lack an α-hydrogen atom on the hydroxyl-bearing carbon atom and cannot dehydrogenate. Instead, they undergo catalytic **dehydration** across adjacent $C-C$ bonds to produce alkenes. 2-Methylpropan-2-ol yields **2-methylpropene (isobutylene)** and water. **Answer: (C)**

Q4. The correct decreasing order of acid strength among the following substituted phenols and water is:

(A) Picric acid $>$ *p*-Nitrophenol $>$ *o*-Nitrophenol $>$ Phenol $>$ Water **(B)** Water $>$ Phenol $>$ *o*-Nitrophenol $>$ *p*-Nitrophenol $>$ Picric acid
(C) Phenol $>$ *p*-Nitrophenol $>$ *o*-Nitrophenol $>$ Picric acid $>$ Water **(D)** *p*-Nitrophenol $>$ Picric acid $>$ *o*-Nitrophenol $>$ Phenol $>$ Water

Sol. Q4. Electron-withdrawing nitro groups ($-NO_2$) stabilize the phenoxide ion through $-I$ and $-M$ resonance effects. Picric acid (2,4,6-trinitrophenol) has three $-NO_2$ groups and is strongly acidic ($pK_a = 0.71$), stronger than carboxylic acids. *p*-Nitrophenol ($pK_a = 7.15$) is slightly more acidic than *o*-nitrophenol ($pK_a = 7.23$) due to absence of intramolecular H-bonding. Phenol ($pK_a \approx 10$) is 10^6 times more acidic than water ($pK_a = 15.7$). Thus, Picric acid $>$ *p*-Nitrophenol $>$ *o*-Nitrophenol $>$ Phenol $>$ Water. **Answer: (A)**

Q5. In the Reimer-Tiemann reaction of phenol with chloroform in aqueous sodium hydroxide, the electrophilic reactive intermediate that attacks the benzene ring is:

(A) Dichloromethyl cation ($CHCl_2^+$) **(B)** Trichloromethyl anion (CCl_3^-)
(C) Chlorocarbene ($:CH-Cl$) **(D)** Dichlorocarbene ($:CCl_2$)

Sol. Q5. In the Reimer-Tiemann reaction, chloroform undergoes α-elimination by hydroxide base to form the neutral, electron-deficient sextet intermediate **dichlorocarbene ($:CCl_2$)**, which acts as the active electrophile attacking the hyperactivated phenoxide ring at the ortho position to yield salicylaldehyde. **Answer: (D)**

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. When phenol is heated with carbon dioxide (CO_2) at 400 K under $4-7$ atm pressure in the presence of sodium hydroxide, followed by acidification, the major product formed is:

(A) Salicylaldehyde **(B)** Benzoic acid
(C) Phthalic acid **(D)** Salicylic acid

Sol. Q6. This is Kolbe's Reaction. Sodium phenoxide undergoes electrophilic aromatic substitution with the weak electrophile carbon dioxide (CO_2) at 400 K and $4-7$ atm. Intramolecular sodium chelation directs carboxylation almost exclusively to the ortho position. Subsequent mineral acidification yields **Salicylic acid (2-hydroxybenzoic acid)**, the fundamental precursor for aspirin. **Answer: (D)**

Q7. A mixture of ortho-nitrophenol and para-nitrophenol can be cleanly separated by steam distillation because:

(A) ortho-Nitrophenol possesses intramolecular hydrogen bonding and is steam-volatile **(B)** para-Nitrophenol possesses intramolecular hydrogen bonding and has lower boiling point
(C) ortho-Nitrophenol forms intermolecular hydrogen bonds with water molecules **(D)** para-Nitrophenol has a lower molecular mass than ortho-nitrophenol

Sol. Q7. ortho-Nitrophenol undergoes **intramolecular hydrogen bonding** (chelation) between $-OH$ and adjacent $-NO_2$, preventing association with other molecules, lowering its boiling point, and making it steam-volatile. Conversely, para-nitrophenol forms extensive **intermolecular hydrogen bonding** leading to molecular association, higher boiling point, and non-volatility in steam. **Answer: (A)**

Q8. Which of the following reaction sequences is the MOST suitable for preparing tert-butyl ethyl ether via Williamson ether synthesis?

(A) Sodium tert-butoxide + Ethyl bromide **(B)** Sodium ethoxide + tert-Butyl bromide
(C) tert-Butyl alcohol + Ethanol with conc. H_2SO_4 at 413 K **(D)** Potassium ethoxide + tert-Butyl chloride at 450 K

Sol. Q8. Williamson ether synthesis involves nucleophilic substitution (S_N2) by an alkoxide ion on an alkyl halide. Alkoxide ions are powerful Brønsted bases. If a tertiary alkyl halide (such as tert-butyl bromide) is used, elimination (E_2) predominates 100%, yielding 2-methylpropene instead of ether. To synthesize tert-butyl ethyl ether, the substrate must be a primary alkyl halide: **Sodium tert-butoxide ($(CH_3)_3C-O^-Na^+$) + Ethyl bromide (C_2H_5 Br)**. **Answer: (A)**

Q9. When anisole (methoxybenzene) is cleaved by heating with concentrated hydriodic acid (HI), the final organic products obtained are:

(A) Iodobenzene and Methanol **(B)** Benzene and Methyl iodide
(C) Phenol and Iodomethane **(D)** Phenol and Methyl iodide

Sol. Q9. In anisole ($C_6H_5-O-CH_3$), the oxygen lone pair is delocalised into the aromatic ring (+R effect), imparting partial double-bond character to the sp^2 $C_{aryl}-O$ bond (136 pm). The sp^3 $C_{alkyl}-O$ bond (142 pm) is longer and weaker. Protonation forms the phenyloxonium ion; iodide ion attacks the less sterically hindered methyl carbon via S_N2 displacement, yielding **Phenol** and **Methyl iodide (CH_3 I)**. **Answer: (D)**

Q10. An organic compound (A) with molecular formula $C_4H_{10}O$ does not react with sodium metal, does not decolorize bromine water, and on heating with excess HI yields only one alkyl halide (B). Compound (A) is:

(A) Butan-1-ol **(B)** Diethyl ether
(C) 1-Methoxypropane **(D)** 2-Methoxypropane

Sol. Q10. Absence of reaction with sodium metal confirms the absence of an active hydroxyl group (not an alcohol), indicating an ether. Since it yields only ONE type of alkyl halide upon heating with excess HI, the ether must be symmetrical. Diethyl ether ($C_2H_5-O-C_2H_5$, $C_4H_{10}O$) undergoes cleavage to produce two equivalents of **ethyl iodide (C_2H_5 I)**. **Answer: (B)**

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. Cleavage of tert-butyl methyl ether with one equivalent of concentrated hydriodic acid (HI) at room temperature produces:

(A) tert-Butyl alcohol and Methyl iodide
(B) tert-Butyl iodide and Methanol
(C) 2-Methylpropene and Methanol
(D) tert-Butyl iodide and Methyl iodide

Sol. Q11. Unlike primary and secondary ethers which cleave via S_N2, ethers containing a tertiary alkyl group undergo cleavage via an S_N1 mechanism. Protonation of the ether oxygen is followed by heterolytic cleavage to generate the exceptionally stable tertiary carbocation ($(CH_3)_3C^+$). Attack by iodide ion on this carbocation yields **tert-butyl iodide ($(CH_3)_3C-I$)** and **methanol (CH_3OH)**. **Answer: (B)**

Q12. The experimental $C-O-C$ bond angle in dimethyl ether (111.7°) is significantly greater than the standard tetrahedral angle (109.5°). This observed widening is due to:

(A) sp^2 hybridisation of the central ethereal oxygen atom
(B) Back-bonding between oxygen lone pairs and empty carbon d-orbitals
(C) Steric repulsion between the two bulky alkyl (methyl) groups
(D) Intramolecular hydrogen bonding between adjacent hydrogen atoms

Sol. Q12. Although the oxygen atom in ethers is sp^3-hybridised with two non-bonding lone pairs, the steric hindrance and electrostatic electron-cloud repulsion between the two bulky alkyl groups attached to oxygen push them apart, expanding the $C-O-C$ bond angle from the tetrahedral value of 109.5° to 111.7°. **Answer: (C)**

Q13. Phenol reacts with aqueous bromine water ($\frac{Br_2}{H_2}$ O) and with bromine in carbon disulphide ($\frac{Br_2}{CS_2}$) at 273 K to give respectively:

(A) o-Bromophenol and 2,4,6-tribromophenol
(B) 2,4,6-Tribromophenol (white ppt) and p-bromophenol (major)
(C) m-Bromophenol and o-bromophenol
(D) p-Bromophenol and 2,4,6-tribromophenol

Sol. Q13. In aqueous medium, water facilitates extensive ionisation of phenol into phenoxide ion ($C_6H_5O^-$), which hyperactivates the benzene ring to undergo polybromination, precipitating 2,4,6-tribromophenol as a white solid. In non-polar CS_2, phenol does not ionise; its ring is less activated, undergoing monobromination to yield p-bromophenol as the major product. **Answer: (B)**

Q14. In the commercial manufacturing process of Phenol from Cumene (isopropylbenzene), the valuable industrial co-product formed in equimolar amounts alongside phenol is:

(A) Acetaldehyde
(B) Benzoic acid
(C) Acetone (Propan-2-one)
(D) Ethyl methyl ketone

Sol. Q14. In the cumene process, cumene is air-oxidized to cumene hydroperoxide, which undergoes acid-catalysed rearrangement with dilute sulphuric acid to cleave into equimolar amounts of **Phenol** and **Acetone (propan-2-one)**: $C_6H_5-C(CH_3)_2-OOH \xrightarrow{H^+} C_6H_5OH + CH_3COCH_3$. The simultaneous recovery of acetone makes this the world's primary industrial process. **Answer: (C)**

Q15. Commercial ethanol is intentionally converted into 'Denatured Alcohol' (Methylated Spirit) before industrial sale by adding:

(A) Pure glucose and sodium bicarbonate to induce crystallization
(B) Methanol (5%), pyridine (foul odour), and copper sulphate (blue colour)
(C) Dilute sulphuric acid and potassium permanganate indicator
(D) Phenolphthalein indicator and concentrated ammonia solution

Sol. Q15. To prevent the illicit diversion and consumption of duty-free industrial alcohol, commercial ethanol is 'denatured' (rendered poisonous and foul-tasting) by blending it with toxic methanol (approx. 5%), pyridine (imparting an intolerable noxious odour), and copper sulphate (giving a warning blue colouration). **Answer: (B)**

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